

Michael Zingale / Publications and Talks

Refereed Publications

80. *AMReX-Astrophysics Microphysics: A set of microphysics routines for astrophysical simulation codes based on the AMReX library*, AMReX-Astro Microphysics Team, K. Bhargava, A. Bishop, Z. Chen, D. Fan, C. Fields, A. M. Jacobs, E. T. Johnson, M. P. Katz, M. Krumholz, C. Malone, A. Nonaka, P. Sharda, A. Smith Clark, F. Timmes, B. Wibking, D. E. Willcox, & M. Zingale, 2025, submitted to Journal of Open Source Software
79. *Multi-dimensional Models of Mixed H/He Flames in X-ray Bursts*, E. T. Johnson & M. Zingale, 2025, submitted to Open Journal of Astrophysics
78. *Multidimensional Nova Simulations with an Extended Buffer and Lower Initial Mixing Temperatures*, A. Smith Clark & M. Zingale, 2025, Open Journal of Astrophysics, volume 8
DOI: 10.33232/001c.136890
77. *Approximating Convective Urca Cooling in a Simmering White Dwarf*, B. Boyd, A. C. Calder, D. Townsley, & M. Zingale, 2025, Journal of Physics: Conference Series, 2997, 1, 012006
DOI: 10.1088/1742-6596/2997/1/012006
76. *The Challenges of Modeling Astrophysical Reacting Flows*, M. Zingale, K. Bhargava, R. Brady, Z. Chen, S. Guichandut, E. T. Johnson, M. Katz, & A. Smith Clark, 2025, Journal of Physics: Conference Series, 2997, 1, 012007
DOI: 10.1088/1742-6596/2997/1/012007
75. *3D Convective Urca Process in a Simmering White Dwarf*, B. Boyd, A. C. Calder, D. Townsley, & M. Zingale, 2025, ApJ, 979, 216
DOI: 10.3847/1538-4357/ad9bb0
74. *Strong Coupling of Hydrodynamics and Reactions in Nuclear Statistical Equilibrium for Modeling Convection in Massive Stars*, M. Zingale, Z. Chen, E. T. Johnson, M. P. Katz, & A. Smith Clark, 2024, ApJ, 977, 30
DOI: 10.3847/1538-4357/ad8a66
73. *Hydrodynamical simulations of proton ingestion flashes in Type I X-ray Bursts*, S. Guichandut, M. Zingale, & A. Cumming, 2024, ApJ, 975, 250
DOI: 10.3847/1538-4357/ad81f7
72. *Sensitivity of Simulations of Double Detonation Type Ia Supernova to Integration Methodology*, M. Zingale, Z. Chen, M. Rasmussen, A. Polin, M. Katz, A. Smith Clark, & E. T. Johnson, 2024, ApJ, 966, 150
DOI: 10.3847/1538-4357/ad3441
71. *Simulating Lateral H/He Flame Propagation in Type I X-ray Bursts*, E. Johnson & M. Zingale, 2024, Journal of Physics: Conference Series, 2742, 1, 012005
DOI: 10.1088/1742-6596/2742/1/012005
70. *pynucastro 2.1: an update on the development of a python library for nuclear astrophysics*, A. Smith Clark, E. T. Johnson, Z. Chen, K. Eiden, M. Zingale, B. Boyd, P. Johnson, & L. Rangel DaCosta, 2024, Journal of Physics: Conference Series, 2742, 1, 012003
DOI: 10.1088/1742-6596/2742/1/012003

69. *Sensitivity of 3D Convective Urca Simulations to Changes in Urca Reactions*, B. Boyd, A. Smith Clark, A. C. Calder, D. M. Townsley, & M. Zingale, 2024, Journal of Physics: Conference Series, 2742, 1, 012001
DOI: 10.1088/1742-6596/2742/1/012001
68. *A Framework for Exploring Nuclear Physics Sensitivity in Numerical Simulations*, Z. Chen, E. Johnson, M. Katz, A. Smith Clark, B. Boyd, & M. Zingale, 2024, Journal of Physics: Conference Series, 2742, 1, 012021
10.1088/1742-6596/2742/1/012021
67. *MESA-Web: A cloud resource for stellar evolution in astronomy curriculum*,
C. E. Fields, R. H. D. Townsend, A. L. Dotter, & M. Zingale, 2023, Astronomy Education Journal, 3, 1, 047ra
DOI: 10.32374/AEJ.2023.3.1.047ra
66. *Sensitivity of He Flames in X-ray Bursts to Nuclear Physics*,
Z. Chen, M. Zingale, & K. Eiden, 2023, ApJ, 955, 128
DOI: 10.3847/1538-4357/acec72
65. *Comparing Early Evolution of Flames in X-ray Bursts in Two and Three Dimensions*,
M. Zingale, K. Eiden, & M. Katz, 2023, ApJ, 952, 160
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64. *pynucastro: A Python Library for Nuclear Astrophysics*,
A. Smith Clark, E. T. Johnson, Z. Chen, K. Eiden, D. E. Willcox, B. Boyd, L. Cao, C. J. DeGrendele, & M. Zingale, 2023, ApJ, 947, 65
DOI: 10.3847/1538-4357/acbaff
63. *Neural Networks for Nuclear Reactions in MAESTROeX*,
D. Fan, D. E. Willcox, C. DeGrendele, M. Zingale, & A. Nonaka, 2022, ApJ, 940, 134
DOI: 10.3847/1538-4357/ac9a4b
62. *An Improved Method for Coupling Hydrodynamics with Astrophysical Reaction Networks*,
M. Zingale, M. P. Katz, A. Nonaka, & M. Rasmussen, 2022, ApJ, 936, 6
DOI: 10.3847/1538-4357/ac8478
61. *Dynamics of Laterally Propagating Flames in X-ray Bursts. II. Realistic Burning & Rotation*,
A. Harpole, N. M. Ford, K. Eiden, M. Zingale, A. D. Willcox, Y. Cavecchi, & M. P. Katz, 2021, ApJ, 912, 36
DOI: 10.3847/1538-4357/abee87
60. *CASTRO: A Massively Parallel Compressible Astrophysics Simulation Code*,
A. Almgren, M. Barrios Sazo, J. Bell, A. Harpole, M. Katz, J. Sexton, D. Willcox, W. Zhang, & M. Zingale, 2020, Journal of Open Source Software, 5, 54, 2513
DOI: 10.21105/joss.02513
59. *Preparing Nuclear Astrophysics for the Exascale*,
M. Katz, A. Almgren, M. Barrios Sazo, K. Eiden, K. Gott, A. Harpole, J. Sexton, D. Willcox, W. Zhang, & M. Zingale, 2020, SC '20: Proceedings of the International Conference for High

- Performance Computing, Networking, Storage and Analysis, 91, 1
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58. *Dynamics of Laterally Propagating Flames in X-ray Bursts. I. Burning Front Structure*,
K. Eiden, M. Zingale, A. Harpole, D. Willcox, Y. Cavecchi, & M. P. Katz, 2020, ApJ, 894, 6
DOI: 10.3847/1538-4357/ab80bc
57. *The Castro AMR Simulation Code: Current and Future Developments*,
M. Zingale, A. S. Almgren, M. Barrios Sazo, J. B. Bell, K. Eiden, A. Harpole, M. P. Katz,
A. J. Nonaka, D. E. Willcox, & W. Zhang, 2020, Journal of Physics: Conference Series, 1623,
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DOI: 10.1088/1742-6596/1623/1/012021
56. *Modelling low Mach number stellar hydrodynamics with MAESTROeX*
A. Harpole, D. Fan, M. P. Katz, A. J. Nonaka, D. E. Willcox, & M. Zingale, 2020, Journal of
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55. *MAESTROeX: A Massively Parallel Low Mach Number Astrophysical Solver*, D. Fan, A. Nonaka,
A. S. Almgren, A. Harpole, & M. Zingale, 2019, ApJ, 887, 212
DOI: 10.3847/1538-4357/ab4f75
54. *Improved Coupling of Hydrodynamics and Nuclear Reactions via Spectral Deferred Corrections*, M.
Zingale, M. P. Katz, J. B. Bell, M. L. Minion, A. J. Nonaka, & W. Zhang, 2019, ApJ, 886, 105
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53. *MAESTROeX: A Massively Parallel Low Mach Number Astrophysical Solver*,
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52. *AMReX: a framework for block-structured adaptive mesh refinement*,
W. Zhang, A. Almgren, V. Beckner, J. Bell, J. Blaschke, C. Chan, M. Day, B. Friesen, K. Gott,
D. Graves, M. P. Katz, A. Myers, T. Nguyen, A. Nonaka, M. Rosso, S. Williams, & M. Zingale,
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51. *Numerical Stability of Detonations in White Dwarf Simulations*,
M. P. Katz & M. Zingale, 2019, ApJ, 874, 169
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50. *pyro: a framework for hydrodynamics explorations and prototyping*,
A. Harpole, M. Zingale, I. Hawke, & T. Chegini, 2019, Journal of Open Source Software, 4, 34,
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49. *Toward Resolved Simulations of Burning Fronts in Thermonuclear X-ray Bursts*,
M. Zingale, K. Eiden, Y. Cavecchi, A. Harpole, J. B. Bell, M. Chang, I. Hawke, M. P. Katz,
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48. *Thermonuclear (Type Ia) Supernovae and Progenitor Evolution*,
A. C. Calder, D. E. Willcox, C. J. DeGrendele, D. Shangase, M. Zingale, & D. M. Townsley,
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47. *Observatory science with eXTP*,
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46. *Turbulence-driven thermal and kinetic energy in the atmospheres of hot Jupiters*,
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45. *pynucastro: an interface to nuclear reaction rates and code generator for reaction network equations*,
D. E. Willcox & M. Zingale, 2018, *Journal of Open Source Software*, 3 (23), 588
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44. *Meeting the Challenges of Modeling Astrophysical Thermonuclear Explosions: Castro, Maestro, and the AMReX Astrophysics Suite*,
M. Zingale, A. S. Almgren, M. G. Barrios Sazo, V. E. Beckner, J. B. Bell, B. Friesen, A. M. Jacobs,
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43. *Best Practices in Running Collaborative GPU Hackathons: Advancing Scientific Applications with a Sustained Impact*,
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42. *Toward Simulating Black Widow Binaries with Castro*,
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41. *Review: White paper on nuclear astrophysics and low energy nuclear physics Part 1: Nuclear astrophysics*,
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C. Brune, A. Champagne, A. Chieffi, A. Couture, P. Danielewicz, R. Diehl, M. El-Eid, J. Escher,
B. Fields, C. Frohlich, F. Herwig, W. R. Hix, C. Iliadis, W. Lynch, G. McLaughlin, B. Meyer, A.
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40. *Low Mach Number Modeling of Convection in Helium Shells on Sub-Chandrasekhar White Dwarfs II: Bulk Properties of Simple Models*,
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39. *Double White Dwarf Mergers on Adaptive Meshes I. Methodology and Code Verification*,
M. P. Katz, M. Zingale, A. C. Calder, F. D. Swesty, A. S. Almgren, W. Zhang, 2016, *ApJ*, 819, 94
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38. *Comparisons of Two- and Three-Dimensional Convection in Type I X-ray Bursts*
M. Zingale, C. M. Malone, A. Nonaka, A. S. Almgren, & J. B. Bell, 2015, *ApJ*, 807, 60
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37. *On the Piecewise Parabolic Method for Compressible Flow with Stellar Equations of State*,
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36. *pyro: A teaching code for computational astrophysical hydrodynamics*,
M. Zingale, 2014, *Astronomy & Computing*, 6, 52
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35. *Multidimensional Modeling of Type I X-ray Bursts. II. Two-Dimensional Convection in a Mixed H/He Accretor*,
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34. *The Deflagration Stage of Chandrasekhar Mass Models For Type Ia Supernovae: I. Early Evolution*,
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33. *Low-Mach Number Modeling of Core Convection in Massive Stars*,
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32. *Low Mach Number Modeling of Convection in Helium Shells on Sub-Chandrasekhar White Dwarfs. I. Methodology*,
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31. *High-Resolution Simulations of Convection Preceding Ignition in Type Ia Supernovae Using Adaptive Mesh Refinement*,

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29. *Multidimensional Modeling of Type I X-ray Bursts. I. Two-Dimensional Convection Prior to the Outburst of a Pure He Accretor*,
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28. *CASTRO: A New Compressible Astrophysical Solver. I. Hydrodynamics and Self-Gravity*,
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27. *MAESTRO: An Adaptive Low Mach Number Hydrodynamics Algorithm for Stellar Flows*,
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22. *Propagation of the First Flames in Type Ia Supernovae*,
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21. *Low Mach Number Modeling of Type Ia Supernovae. II. Energy Evolution*,
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20. *Low Mach Number Modeling of Type Ia Supernovae. I. Hydrodynamics*,
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19. *Three-Dimensional Numerical Simulations of Rayleigh-Taylor Unstable Flames in Type Ia Supernovae*,
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18. *Validating an Astrophysical Simulation Codes*,
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17. *Direct Numerical Simulations of Type Ia Supernovae Flames II: The Rayleigh-Taylor Instability*,
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4. *High-Performance Reactive Fluid Flow Simulations Using Adaptive Mesh Refinement on Thousands of Processors*,
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1. *Flash Code: Studying Astrophysical Thermonuclear Flashes,*

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28. *Numerical Treatment of Shock-induced Nuclear Burning in Double Detonation Type Ia Supernovae,*
R. Brady & M. Zingale, 2025, *Research Notes of the AAS*, 9, 5, 113
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27. *Well-Balanced Hydrodynamics for the Piecewise Parabolic Method with Characteristic Tracing,*
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DOI: 10.3847/2515-5172/ad76b0
26. *A Fully Explicit Integrator for Modeling Astrophysical Reactive Flows,*
P. Johnson, M. Zingale, E. T. Johnson, A. Smith, & K. Niemeyer, 2023, *Research Notes of the AAS*, 7, 12, 282
DOI: 10.3847/2515-5172/ad175d
25. *Practical Effects of Integrating Temperature with Strang Split Reactions,*
M. Zingale, M. P. Katz, D. E. Willcox, & A. Harpole, 2021, *Research Notes of the AAS*, 5, 4, 71
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24. *The LOFT mission concept: a status update,*
M. Feroci et al., 2016, *Proc. SPIE 9905, Space Telescopes and Instrumentation 2016: Ultraviolet to Gamma Ray*, 99051R, July 25, 2016
DOI: 10.48550/arXiv.1501.02776
23. *Understanding Ignition in Type Ia Supernovae,*
M. Zingale, A. Jacobs, A. S. Almgren, J. B. Bell, A. Nonaka, C. Malone, & S. Woosley, 2015, extended abstract for the 25th International Colloquium on the Dynamics of Explosions and Reactive Systems, Leeds, UK, Aug. 2–7, 2015
22. *Low Mach Number Modeling of Stratified Flows,*
A. S. Almgren, J. B. Bell, A. Nonaka, & M. Zingale, 2014, in *Finite Volumes for Complex Applications VII: Methods, Theoretical Aspects—FVCA 7*, Berlin, June 2014, ed. Fuhrmann, J., Ohlberger, M., & Rohde, C., 3–15; *Proceedings of the FVCA7 - The International Symposium of Finite Volumes for Complex Applications VII Berlin*, June 15–20, 2014
21. *From Convection to Explosion: End-to-End Simulation of Type Ia Supernovae,*
A. Nonaka, A. S. Almgren, J. B. Bell, H. Ma, S. E. Woosley, & M. Zingale, 2011, *Proceedings of SciDAC 2011*, Denver, CO, July 10–14, 2011

20. *MAESTRO, CASTRO, and SEDONA — Petascale Codes for Astrophysical Applications*,
A. Almgren, J. Bell, D. Kasen, M. Lijewski, A. Nonaka, P. Nugent, C. Rendlement, R. Thomas,
& M. Zingale, 2010, Proceedings of the 2010 Scientific Discovery through Advanced Comput-
ing (SciDAC) Conference. Chattanooga, Tennessee, July 11–15, 2010. Oak Ridge National
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Conference. Chattanooga, Tennessee, July 11–15, 2010. Oak Ridge National Laboratory.
18. *Type Ia Supernovae: Advances in Large Scale Simulation*,
S. E. Woosley, A. S. Almgren, A. J. Aspden, J. B. Bell, D. Kasen, A. R. Kerstein, H. Ma, A. Nonaka,
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17. *Astrophysical Applications of the Maestro Code*,
M. Zingale, A. S. Almgren, J. B. Bell, C. M. Malone, & A. Nonaka, 2008, Proceedings of SciDAC
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16. *Type Ia supernovae*,
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10. *Investigations of Pointwise Ignition of Helium Deflagrations on Neutron Stars*,
M. Zingale, S. E. Woosley, A. Cumming, A. Calder, L. J. Dursi, B. Fryxell, K. Olson, P. Ricker, R. Rosner, & F. X. Timmes, 2002, 3D Stellar Evolution, ASP Conference Proceedings, Vol. 293, 22–26 July 2002 at UC Davis, Livermore, CA, Ed. by S. Turcotte, S. C. Keller, & R. M. Cavallo.
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L. J. Dursi, A. C. Calder, A. Alexakis, J. W. Truran, M. Zingale, B. Fryxell, P. Ricker, F. X. Timmes, & K. Olson, 2002, Classical Nova Explosions: International Conference on Classical Nova Explosions. AIP Conference Proceedings, Vol. 637. Sitges, Spain, 20–24 May, 2002. Edited by M. Hernanz & J. Jose
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8. *Mixing by Non-linear Gravity Wave Breaking on a White Dwarf Surface*,
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5. *Quenching Processes in Flame-Vortex Interactions*,
M. Zingale, J. C. Niemeyer, F. X. Timmes, L. J. Dursi, A. C. Calder, B. Fryxell, D. Q. Lamb, K. Olson, P. Ricker, R. Rosner, J. W. Truran, & P. MacNeice, 2001, 20th Texas Symposium on Relativistic Astrophysics, Austin, Texas, 10–15 Dec. 2000, Melville, NY: AIP Conference Proceedings, Vol. 586. Edited by J. C. Wheeler & H. Martel, also AIP Conference Series 586, 490–492.
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4. *Simulations of Astrophysical Fluid Instabilities*,
A. C. Calder, B. Fryxell, R. Rosner, L. J. Dursi, K. Olson, P. M. Ricker, F. X. Timmes, M. Zingale, P. MacNeice, & H. M. Tufo, 2001, 20th Texas Symposium on Relativistic Astrophysics, Austin,

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3. *Adaptive Mesh Simulations Of Astrophysical Detonations Using the ASCI Flash Code,*

B. Fryxell, A. C. Calder, L. J. Dursi, D. Q. Lamb, P. MacNeice, K. Olson, P. M. Ricker, R. Rosner, F. X. Timmes, J. W. Truran, H. M. Tufo, & M. Zingale, Proceedings of the VII International Workshop on Advanced Computing and Analysis Techniques in Physics Research (ACAT 2000), Fermilab, October 16–20, 2000.

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2. *Large-Scale Simulations of Clusters of Galaxies,*

P. M. Ricker, A. C. Calder, L. J. Dursi, B. Fryxell, D. Q. Lamb, P. MacNeice, K. Olson, R. Rosner, F. X. Timmes, J. W. Truran, H. M. Tufo, & M. Zingale, Proceedings of the VII International Workshop on Advanced Computing and Analysis Techniques in Physics Research (ACAT 2000), Fermilab, October 16–20, 2000.

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1. *Helium Detonations on Neutron Stars,*

B. Fryxell, M. Zingale, F. X. Timmes, D. Q. Lamb, K. Olson, A. C. Calder, L. J. Dursi, P. Ricker, R. Rosner, J. W. Truran, P. MacNeice, & H. Tufo, Proceedings of the 10th Workshop on “Nuclear Astrophysics”, Ringberg Castle, Tegernsee, Germany, March 20–25 2000.

Open Books

2. *Teaching and Learning with Jupyter,*

L. A. Barba, L. J. Barker, D. S. Blank, J. Brown, A. B. Downey, T. George, L. J. Heagy, K. T. Mandli, J. K. Moore, D. Lippert, K. E. Niemeyer, R. R. Watkins, R. H. West, E. Wickes, C. Willing, & M. Zingale

<https://jupyter4edu.github.io/jupyter-edu-book/>

1. *Introduction to Computational Astrophysical Hydrodynamics,*

M. Zingale

https://github.com/Open-Astrophysics-Bookshelf/numerical_exercises

White Papers

8. *Multidisciplinary Science in the Multimessenger Era,*

E. Burns et al., 2025, White Paper of the Third Time-Domain And Multimessenger Astrophysics workshop: Multidisciplinary Science in the Multimessenger Era

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7. *Horizons: Nuclear Astrophysics in the 2020s and Beyond,*

H. Schatz et al., 2022, Journal of Physics G Nuclear Physics, 49, 11, 110502

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6. *MMA SAG: Thermonuclear Supernovae*,
M. Zingale, C. Fryer, A. Hungerford, S. Safi-Harb, R. Trappitsch, R. Fisher, A. Calder, & K. Shen, *Astro2020: Decadal Survey on Astronomy and Astrophysics*, science white papers, no. 259; *Bulletin of the American Astronomical Society*, Vol. 51, Issue 3, id. 259 (2019)
5. *Nuclear Physics Exascale Requirements Review: An Office of Science review sponsored jointly by Advanced Scientific Computing Research and Nuclear Physics*,
J. Carlson et al. 2017, US Department of Energy, Washington, DC (United States). *Advanced Scientific Computing Research and Nuclear Physics*
4. *The Importance of Computation in Astronomy Education*,
M. Zingale, F. X. Timmes, R. Fisher, & B. W. O'Shea, 2016, white paper submitted to the AAS Education Taskforce call
(<https://aas.org/posts/opportunity/2016/04/aas-task-force-education-begins-its-work>)
3. *White Paper on Nuclear Astrophysics*,
A. Arcones et al. 2016, Community white paper based on 2012 JINA Town Meeting in Detroit, MI, and 2014 APS Town Meeting in College Station, TX
2. *Modeling Astrophysical Explosions with Sustained Exascale Computing*,
M. Zingale, A. C. Calder, C. M. Malone, & F. X. Timmes, 2015, Response to RFI NOT-GM-15-122: *Science Drivers Requiring Capable Exascale High Performance Computing*
1. *The LOFT perspective on neutron star thermonuclear bursts*,
J. J. M. in 't Zand, D. Altamirano, D. R. Ballantyne, S. Bhattacharyya, E. F. Brown, Y. Cavecchi, D. Chakrabarty, J. Chenevez, A. Cumming, N. Degenaar, M. Falanga, D. K. Galloway, A. Heger, J. José, L. Keek, M. Méndez, S. Mahmoodifar, M. Linares, C. M. Malone, M. C. Miller, F. B. S. Paerels, J. Poutanen, A. Różańska, H. Schatz, M. Serino, V. F. Suleimanov, T. E. Strohmayer, F.-K. Thielemann, A. L. Watts, N. N. Weinberg, S. E. Woosley, W. Yu, S. Zhang, & M. Zingale, 2015, White Paper in Support of the Mission Concept of the Large Observatory For x-ray Timing

Invited Lectures / Seminars / Colloquia

- | | |
|------------|---|
| 05/06/2025 | Invited Keynote at Cray Users Group 2025 meeting, Jersey City, NJ, <i>What I Learned About Supercomputing from Blowing Up Stars</i> |
| 10/09/2024 | Invited speaker at APS-DNP 2024 meeting, Mini-Symposium: Nuclear Data in the Cosmos, <i>pynucastro: A python library for connecting nuclear data to astrophysical simulations</i> |
| 09/13/2024 | IRENA Online Seminar Series, <i>pynucastro: A python library connecting nuclear experiment data and astrophysics</i> |
| 08/07/2024 | Invited talk at the Rise_Time conference, Purdue University, <i>Modeling Astrophysical Reacting Flows</i> |
| 07/30/2024 | Modeling Talk seminar series at Google research / x.company, <i>The Challenges of Modeling Stellar Explosions</i> |

- 07/04/2024 Invited talk at AstroNum 2024, La Rochelle, France, *Strong Coupling of Hydrodynamics and Reactions in Astrophysical Flows*
- 04/15/2024 Colloquium at Temple University, *Modeling Stellar Explosions*
- 03/06/2024 Invited minisymposium talk in the Performance Optimization for Multiphysics Applications on Structured Mesh with Particles minisymposium at the 2024 SIAM Conference on Parallel Processing for Scientific Computing, *Modeling Astrophysical Reactive Flows with Castro at the Exascale*
- 11/04/2023 Invited talk at APS-MAS, University of Delaware, *Engaging Students in Computational (Astro)Physics Courses and Research*
- 10/23/2023 Invited talk at the Interfaces and Mixing, Exploration Conference, 23-26 Oct 2023, Kavli Institute for Theoretical Physics, *Convective Mixing in Stellar Reactive Flows*
- 06/28/2023 Invited talk at AstroNum 2023–15th International Conference on Numerical Modeling of Space Plasma Flows, Pasadena, CA, USA, *Coupling Hydrodynamics and Astrophysical Reaction Networks*
- 11/21/2022 Invited talk in *Astrophysical Turbulence: Current Understanding and Modeling Challenges* minisymposium at the APS/Division of Fluid Dynamics 2022 meeting, *Modeling Astrophysics Reactions in Turbulent Environments*
- 11/14/2022 Invited talk a HPC session of the Computing Working Group at the Argonne Town Meeting on Nuclear Structure, Reactions, and Astrophysics, *Blowing Up Stars Over the Next Decade*
- 07/28/2022 Universiti Malaya (Kuala Lumpur, Malaysia), Department of Physics Colloquium, *Modeling Astrophysical Thermonuclear Explosions*
- 05/20/2022 FastMath seminar, *Algorithmic Improvements for Coupling Hydrodynamics and Reactions in Astrophysical Flows*
- 05/04/2022 Invited speaker in the AMReX Breakout Session of the 2022 ECP Annual Meeting, *Castro Developments for Exascale Platforms*
- 03/16/2022 Invited Focus Session Speaker in *Extreme-Scale Computational Science Discovery in Fluid Dynamics and Related Disciplines*, APS March Meeting, *The Challenges of Modeling Astrophysical Reactive Flows*
- 11/17/2021 Invited panelist for KITP Transport in Stars workshop on convection (KITP, Santa Barbara, CA)
- 10/19/2021 Astronomy Society of Ireland Colloquium, *The Challenges of Modeling Astrophysical Reactive Flows*
- 10/18/2021 Michigan State University, Department of Computational Mathematics, Science and Engineering Colloquium, *The Challenges of Modeling Astrophysical Reactive Flows*
- 03/02/2021 Invited talk in the *Computational Methods in Explosive Nuclear Astrophysics* session at the SIAM Computational Science and Engineering 2021 meeting, *Modeling Astrophysical Reacting Flows*
- 01/30/2020 Princeton University, Department of Astrophysical Sciences Thunch talk, *Modeling Stellar Explosions*

- 01/09/2020 Invited presentation at Texas Advanced Computing Center *Workshop on Future Directions in Extreme Scale Computing for Scientific Grand Challenges on Challenges in Modeling Astrophysical Thermonuclear Explosions*
- 11/06/2019 Stony Brook University, Department of Physics and Astronomy Colloquium, *Modeling Thermonuclear X-ray Bursts*
- 07/05/2019 Invited talk at AstroNum 2019—14th International Conference on Numerical Modeling of Space Plasma Flows, Paris, France, *Improved Coupling of Hydrodynamics and Nuclear Burning in Astrophysical Flows using SDC*
- 02/26/2019 Invited talk in the *Spectral Deferred Correction Methods for Temporal Integration* session at the SIAM Computational Science and Engineering 2019 meeting, *Improved Coupling of Hydrodynamics and Nuclear Burning in Astrophysical Flows using SDC*
- 10/12/2018 Flatiron Institute Center for Computational Astrophysics Colloquium, *Algorithmic Demands for Modeling X-ray Bursts and Type Ia Supernovae*
- 08/23/2018 Talk at the TEAMS Collaboration meeting, *StarKiller Microphysics*
- 06/26/2018 Invited talk at AstroNum 2018—13th International Conference on Numerical Modeling of Space Plasma Flows, Panama City, Florida, *Modeling X-ray Bursts with the AMReX Astrophysics Suite*
- 08/10/2017 Seminar at LLNL High Energy Density Science Center, LLNL, *Modeling Stellar Explosions with the AMReX Astrophysics Suite*
- 07/27/2017 Seminar at Computational Science Initiative, BNL, *The AMReX Astrophysics Suite: Simulating the Stars at the Exascale*
- 06/30/2017 Invited talk at AstroNum 2017—12th International Conference on Numerical Modeling of Space Plasma Flows, St. Malo, France, *Computational Challenges of Modeling X-ray Bursts and Type Ia Supernovae*
- 06/02/2017 Invited participant / overview talk at Stellar Hydro Days, University of Victoria, *Modeling Stellar Convection and Explosions with Maestro, Castro, and the BoxLib/AMReX Astrophysics Suite*
- 04/05/2017 Astronomy Seminar at Michigan State University, *Computational Challenges of Modeling X-ray Bursts and Type Ia Supernovae*
- 02/23/2017 Seminar at Stony Brook Institute for Advanced Computational Science, *Computational Challenges of Modeling X-ray Bursts and Type Ia Supernovae*
- 06/15/2016 Case study talk at DOE Nuclear Physics / ASCR Exascale Requirements Review, Gaithersburg, MD, *Thermonuclear Transients*
- 04/29/2016 Seminar at Oak Ridge National Laboratory, *Modeling Stellar Explosions with Maestro, Castro, and the BoxLib Astrophysics Suite*
- 03/17/2016 Talk at the 18th Workshop on Nuclear Astrophysics, Ringberg Castle, Tegernsee, Germany, *Models of convection in X-ray bursts and pre-SNe Ia white dwarfs*
- 02/26/2016 Seminar at the U. S. Naval Research Laboratory, *Computational Challenges of Modeling X-ray Bursts and Type Ia Supernovae*
- 08/02/2015 Invited talk at the *International Colloquium on the Dynamics of Explosions and Reactive Systems (ICDERS)*, Leeds, UK, *Understanding Ignition in Type Ia Supernovae*

- 06/22/2015 Invited talk at the *OLCF User's Meeting*, ORNL, Oak Ridge, TN, *Computation Challenges of Modeling Astrophysical Explosions*
- 06/03/2015 Invited talk at the *Fifty One Ergs* meeting, NCSU, *Modeling the Early Phases of Type Ia Supernovae*
- 05/24/2015 "Setting the Stage" talk on *Stellar Hydrodynamics* at the JINA GNASH: *The anomalous metal-poor stars and convective-reactive nuclear astrophysics* workshop, Victoria, BC, Canada, <http://jina-cee.phys.uvic.ca/gnash-workshop/talks-and-contributions/monday/setting-the-stage>
- 04/08/2015 Seminar at U Mass Dartmouth, *Algorithmic Developments for Modeling Stellar Explosions*
- 01/15/2015 CCS-2 Seminar at Los Alamos National Laboratory, *The Challenges of Modeling Type Ia Supernovae and X-ray Bursts*
- 09/15/2014 Invited talk at the *Type Ia Supernovae: progenitors, explosions, and cosmology* conference, Chicago, IL, *Modeling the Early Phases of SNe Ia*, <https://kicp-workshops.uchicago.edu/sn2014/presentations.php>
- 04/30/2014 Invited presentation at *Large Scale Computing and Storage Requirements for Nuclear Physics (NP): Target 2017 meeting*, *Convection in X-ray Bursts*
- 02/28/2014 Astronomy Seminar at the Center for Cosmology and Particle Physics, New York University, *Modeling Convective Burning in Type Ia Supernovae and X-ray Bursts*
- 09/27/2013 Nuclear Theory Seminar at Brookhaven National Lab, *Modeling Convective Burning in Type Ia Supernovae and X-ray Bursts*
- 07/09/2013 Seminar at the Flash Center, University of Chicago, *Modeling Convective Burning in Type Ia Supernovae and X-ray Bursts*
- 10/10/2012 Astro Computation working group at *2012 Nuclear Astrophysics Town Meeting*, *Thermonuclear Driven Events*
- 04/04/2012 Nuclear Astrophysics Seminar at Ohio University entitled *The Challenges of Modeling Explosive Phenomena*
- 07/28/2010 Invited talk at the Lorentz Center Workshop on *X-ray Bursts and Burst Oscillations* entitled *The Algorithmic Challenges of Multidimensional Models of X-ray Bursts*, <http://www.lorentzcenter.nl/lc/web/2010/408/info.php3?wsid=408>
- 05/13/2010 Joint NRAO / UVa Dept. of Astronomy Colloquium (Charlottesville, VA) entitled *Modeling Convection and Ignition in Type Ia Supernovae*
- 03/31/2010 Center for the Study of Cosmic Evolution Seminar, Dept. of Physics and Astronomy, Michigan State University (E. Lansing, MI), entitled: *Modeling Convection and Ignition in Type Ia Supernovae*
- 05/12/2009 Astronomy Seminar at the American Museum of Natural History (New York, NY), entitled: *Modeling Convection and Ignition in Type Ia Supernovae*
- 09/30/2008 Astronomy Seminar at the Institute for Advanced Studies (Princeton, NJ), entitled: *New Methods for Modeling Type Ia Supernovae*

- 07/15/2008 Invited Poster at the *SciDAC 2008* conference (Seattle, WA), entitled: *Astrophysical Applications of the Maestro Code* (with co-authors: A. S. Almgren, J. B. Bell, C. M. Malone, & A. J. Nonaka)
- 04/06/2007 Astronomy Seminar at Rutgers University (New Brunswick, NJ), entitled: *The Challenges of Modeling Type Ia Supernova*
- 10/31/2006 Astronomy Colloquia at McGill University (Montreal, CA), entitled: *Understanding Type Ia Supernovae*
- 06/27/2006 Invited talk at the *SciDAC 2006* conference (Denver, CO), entitled: *The Challenges of Modeling Type Ia Supernovae*
- 10/03/2005 T-13 Seminar, Los Alamos National Laboratory, entitled: *Simulations of Thermonuclear Flames in Type Ia Supernovae*
- 06/26/2005 Invited poster at the *SciDAC 2005* conference (San Francisco, CA), *The Physics of Thermonuclear Flames in Type Ia Supernovae*
- 03/01/2005 Astronomy Seminar at SUNY Stony Brook, *Flame Instabilities in Type Ia Supernovae*
- 02/23/2005 N Division Seminar, Lawrence Livermore National Laboratory, *Flame Instabilities in Type Ia Supernovae*
- 12/17/2003 Astrophysics Seminar, Institute for Advanced Study, Princeton, NJ, *Flame Instabilities in Type Ia Supernovae*

Popular Press Features

How Stars Explode, Forbes.com, Oct. 1, 2009

(<http://www.forbes.com/2009/09/30/supernovae-universe-science-technology-breakthroughs-stars.html>)

Unveiled: The First Full 3-D Model of a Star Going Supernova, Popular Science Online, Sept. 24, 2009

(<http://www.popsci.com/military-aviation-amp-space/article/2009-09/first-3-d-models-white-dwarf-supernova>)

Flash Upon a Neutron Star, American Scientist, Sept.–Oct. 2000, vol. 88, no. 5, p. 400.

Popular Press Mentions

Stars Go Kaboom, Spilling Cosmic Secrets, Science News, 2009, Vol. 176, #4 (Aug. 15, 2009)

(see also http://www.sciencenews.org/view/feature/id/46029/title/Stars_go_kaboom,_spilling_cosmic_secrets)

Supernova explosion simulated in exquisite detail, New Scientist Online, July 2006

(<http://www.newscientist.com/article/dn9604-supernova-explosion-simulated-in-exquisite-detail.html>)

Life-or-Death Question: How Supernovas Happen? NY Times, Nov. 9, 2004.

Physics Today cover, Feb. 2002.